Provenance of 'dirac matrices': where every matrix, equation and result comes from

This provenance record accompanies the proof file 'dirac matrices'. It states, for every object and every claim, its origin (source notebook cell, published reference, or computation) and how it was checked. All equations that are referred to are written out here.

1. Primary source: the notebook Pre-gravity6plane.nb

Author: Patrick L. Nash, (C) 2022, GNU General Public License. File size 23,772,734 bytes, read in full (1,415 styled cells: 650 input, 546 output, 156 text, 32 section, 6 title). The constructions below are copied from the following cells (cell numbers count the styled cells of the notebook in document order).

object	definition (as used)	notebook cells
coordinates	$X = \{x_0, x_1, \ldots, x_7\}$	45
metric	$\eta = \mathrm{diag}(1,1,1,1,-1,-1,-1,-1)$	66, 157
spinor metric	$\sigma = \begin{pmatrix} 0 & I_4 \\ I_4 & 0 \end{pmatrix}$	73
self-dual / anti-self-dual	$(Q^h_a)_{pq} = \epsilon_{hpq4}$, $(Q^h_b)_{pq} = \delta_{p4}\delta_{qh} - \delta_{ph}\delta_{q4}$, $s_h = Q^h_a - Q^h_b$, $t_h = Q^h_a + Q^h_b$	203, 208, 209
six antisymmetric 8x8	$\begin{pmatrix} 0 & s_h \\ s_h & 0 \end{pmatrix}$ $(h = 1,2,3)$, $\begin{pmatrix} 0 & t_h \\ -t_h & 0 \end{pmatrix}$ $(h = 3,2,1)$	213
tau matrices	$\tau_0 = I_8$, $\tau_h =$ the six matrices, $\tau_7 = \tau_1 \cdots \tau_6$, $\bar\tau_A = \sigma(\sigma\tau_A)^T$	265
Dirac matrices	$T^A = (T16^A)[A] = \begin{pmatrix} 0 & \bar\tau_A \\ \tau_A & 0 \end{pmatrix}$	308
chirality	$(T16^A)[8] = T^0T^1\cdots T^7$	309
spinor metric 16	$\sigma_{16} = T^0T^1T^2T^3 = \mathrm{diag}(-\sigma, \sigma)$	306, 333
chiral projectors	$P_L = \frac{1}{2}(I_{16} - (T16^A)[8])$, $P_R = \frac{1}{2}(I_{16} + (T16^A)[8])$	378, 380, 382
scaled commutators	$S^{AB} = \frac{1}{4}(T^AT^B - T^BT^A)$ and its so(4,4) relations	385-391
256-element basis	products of the T^A	453-491
stored numerical results used for comparison	output cells 310 ($(T16^A)[8]$), 312 (σ_{16}), 379 (P_L), 381 (P_R), 494 ($(T16^A)[0..7]$)	310, 312, 379, 381, 494

The notebook also states the defining identity

$$\tfrac{1}{2}\,(\tau_A\bar\tau_B + \tau_B\bar\tau_A) = \eta_{AB}\, I_8$$

(its text cell 68 and the check of cell 265), which is the starting point of the anticommutation proof.

2. Published references quoted by the notebook (used for terminology and for the block form)

- P. L. Nash, "Second gravity", J. Math. Phys. 51, 042501 (2010): block form $t^A = \begin{pmatrix} 0 & \bar\tau^A \\ \tau^A & 0 \end{pmatrix}$ and $t^At^B - t^Bt^A = 4\,\mathrm{diag}(D^{AB}_{(1)}, D^{AB}_{(2)})$ with $4D^{AB}_{(1)} = \bar\tau^A\tau^B - \bar\tau^B\tau^A$, $4D^{AB}_{(2)} = \tau^A\bar\tau^B - \tau^B\bar\tau^A$ (notebook image cells 70, 255, 304).
- E. A. Lord, "The Dirac spinor in six dimensions", Proc. Camb. Phil. Soc. 64, 765-778 (1968): reduced Brauer-Weyl generators (notebook cells 722-746).
- P. A. M. Dirac, "A remarkable representation of the 3 + 2 de Sitter group", J. Math. Phys. 4, 901-909 (1963) (notebook cell 1406).
- P. L. Nash, "Identities satisfied by the generators of the Dirac algebra", J. Math. Phys. 25, 204 (1984): conventions $S^{\alpha\beta} = -\frac{1}{4}[\gamma^\alpha, \gamma^\beta]$ for the 4x4 case (notebook cells 619, 677, 703).

Standard mathematics used in the proof of Theorem 3 (not taken from the notebook):

- Cartan-Dieudonne theorem: every element of the orthogonal group of a non-degenerate real quadratic space of dimension n is a product of at most n reflections $v \mapsto v - 2\frac{(v\cdot n)}{(n\cdot n)}n$.
- Definition of Pin and Spin groups as groups of products of unit vectors in the Clifford algebra and the twisted adjoint action $\rho(g)v = (-1)^{|g|}gvg^{-1}$; see e.g. H. B. Lawson and M.-L. Michelsohn, Spin Geometry, Princeton University Press (1989), Chapter I.
- Classification: the real Clifford algebra of signature (4,4) is the full matrix algebra M(16, R), consistent with the computed rank 256.

3. Computations (all exact)

- Software: Wolfram Language / Mathematica 15.0.1 for Mac OS X ARM (64-bit), run through wolframscript on 2026-10-30.
- Script: `verify_dirac_matrices.wls` (with `core.wl`, which re-types the notebook definitions of Section 1). Every check uses exact integer or exact symbolic arithmetic; no floating point number is involved.
- Log: `verify_dirac_matrices.log`. Results: checks 01-27 (reality; $T^AT^B + T^BT^A = 2\eta^{AB}I_{16}$ for 64 pairs; transposes; squares; chirality; projectors; σ_{16} properties; so(4,4) relations for 4096 quadruples; vector action for 512 triples; rank 28 of the S^{AB}; $(S^{AB})^2 = -\frac{1}{4}\eta^{AA}\eta^{BB}I_{16}$; 12 compact and 16 non-compact generators; closed forms of $e^{\theta S^{AB}}$; $\Lambda\eta\Lambda^T = \eta$, $\det\Lambda = 1$; $e^{2\pi S^{AB}} = -I_{16}$; reflections; four components of O(4,4); rank 256 of the products; trace orthogonality $\mathrm{Tr}(B_I B_J^{-1}) = 16\delta_{IJ}$; block-diagonal S^{AB}; the tau identity) all returned True (or the stated numbers 28, {12,16}, {256,256}, and the component signs {{1,1},{-1,1},{1,-1},{-1,-1}}).
- Comparison with the notebook: checks 28-31 parsed the stored output cells 494, 310, 312, 379, 381 of Pre-gravity6plane.nb with `ToExpression` and compared them with the recomputed matrices: identical.
- Display: the matrices printed in 'dirac matrices' were written by the script `gen_dm.py` directly from the JSON file `dm_data.json` exported by `verify_dirac_matrices.wls`; no matrix was typed by hand.
- Conversion: the LaTeX version is produced from the Markdown version by `md2tex.py` and compiled with pdfLaTeX (TeX Live 2026).

4. Statements that are new in this work

The construction of the matrices is that of Pre-gravity6plane.nb. New here are: the written proofs of Theorems 1-4 (in particular the upper-block argument $\bar\tau_A\tau_B + \bar\tau_B\tau_A = \eta_{AA}X + \eta_{BB}X^{-1} = 0$ with $X = \tau_A^{-1}\tau_B$ for $A \neq B$), the factorization $e^{\theta S^{AB}} = \eta^{AA}T^A(cT^A + \eta^{AA}sT^B)$ into unit vectors, the identification of the four components of O(4,4) reached by I_{16}, T^0, T^4, T^0T^4, and the observation $(T^C)^T\sigma_{16}T^C = -\eta^{CC}\sigma_{16}$ (sign change of $\bar\psi\psi$ under space-like reflections).